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# A Graph Neural Network for Pandemic Forecasting and Planning

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## Abstract

We study the problem of forecasting the spatio-temporal evolution of COVID-19 across U.S. states using fine-grained state-variant-level case data and a transportation-informed notion of connectivity. We construct a variety of models based on a state-level graph with nodes connected via a weighted transit network. We evaluate the model on multi-step-ahead prediction of daily total cases by state over a seven-day horizon, comparing against a variety of naive and machine learning baselines. Empirically, our best specification achieves an RMSE of 2411.11 on held-out test data, improving upon the best baseline (3113.63) by 22.56%. These results suggest that incorporating transportation-based connectivity and individual variant-level disaggregation can provide meaningful gains over purely temporal baselines in forecasting aggregate COVID-19 incidence.

## 1 Introduction

### 1.1 Project Overview

Research on pandemics remains a highly active area, and numerous studies have incorporated graph learning to analyze or predict the spread of infectious diseases. Our goal is to forecast the transmission dynamics of a given disease—specifically COVID-19—across a spatial network of locations. To achieve this, we aim to enhance existing models to make them more expressive and effective, and to develop a new architecture inspired by prior work in this domain.

### 1.2 Literature Review

Traditional epidemic prediction research can be broadly classified into two categories: causal models and statistical models. The first category, causal models, includes compartmental models and agent-based models. These approaches often employ the SIR model or its variants to simulate epidemic spread mechanisms [[11], [14]]. Compartmental models emphasize the mathematical formulation of population-level dynamics. For instance, [22] applied the SIR model within a stochastic framework to study epidemic spread, highlighting that large super-spreading events should be primary targets for intervention. Similarly, [17] used agent-based simulation modeling to demonstrate the importance of social distancing measures and hospital capacity in epidemic control. The second category, statistical models, leverages historical data for prediction. Autoregressive (AR) models, for example, have been widely applied to epidemic forecasting due to their effectiveness in handling time series data. [16] employed a linear autoregressive exogenous (ARX) model based on traditional surveillance data and evaluated its predictive performance. [13] examined the effectiveness of Elastic Net regression and extended it by incorporating query data into a nonlinear regression framework based on composite Gaussian Processes. Finally, they enhanced query-only predictions with an autoregressive component, thereby injecting prior knowledge of disease dynamics into the model.

Various graph learning methods have been proposed for epidemic prediction. For example, [12] employ Graph Neural Networks with mobility data, constructing a single large-scale spatio-temporal

graph in which nodes represent region-level human mobility, spatial edges encode inter-regional connectivity, and temporal edges capture node features over time. [10] design a high-resolution graph that incorporates factors such as mask-wearing willingness, daily infections, and deaths. Their method learns from a multivariate spatio-temporal graph, where nodes represent regions with daily confirmed cases and deaths, and edges reflect inter-regional contacts based on mobility. They also introduce a transmission model with a temporal margin to characterize changes over time, building a COVID-19 model with STGNNs to predict and validate infection spread. [3] develop two types of GNN models: (1) graph theory-based neural networks (GTNNs) and (2) neighborhood-based neural networks (NGNNs). In both graphs, nodes correspond to individual U.S. states. The GTNN edges capture functional connectivity between states, while NGNN edges connect neighboring states. In another direction, [7] integrate prediction and interpretability by presenting a multimodal learning framework that combines statistical regression with machine learning models to predict local COVID-19 cases in Germany. Similarly, [15] leverage graph representation learning to model the relationship between population movement and COVID-19 spread. Their GNN-based approach encodes diffusion patterns into the learning process, predicts future case counts, and applies model-agnostic meta-learning to transfer knowledge across countries, taking advantage of asynchronous outbreak patterns. [18] examine the effectiveness of GNNs in estimating epidemic spreading states, while [24] frame the problem as a spatio-temporal network representation task. They propose STEP, a Spatio-Temporal Epidemic Prediction framework, which estimates a city’s infection risk by integrating diverse real-world conditions (e.g., City Risk Index, climate, medical infrastructure) into graph-structured data. Their model further employs a multi-head attention mechanism to extract implicit features for city-level prediction.

Expanding beyond the state-level, [23] develop a multi-scale GNN by including graphs of cases at the county level and the state level. This approach allows for incorporating diffusion patterns of disease at multiple scales, which is both more granular and representative of real-world disease mechanics.

## 2 Problem Definition

We consider the problem of forecasting the spatio-temporal evolution of COVID-19 across the United States. Specifically, given historical daily case counts by variant at the state level, the objective is to predict future daily case counts over a fixed forecast horizon. A central feature of the problem is that regions are not epidemiologically independent: inter-regional movement along major transportation corridors like highways and air corridors create channels through which outbreaks in one area influence incidence in others. The forecasting task, therefore, requires leveraging not only temporal dynamics within each jurisdiction but also the spatial pattern of connectivity between states.

This setting is further complicated by the empirical characteristics of the data. Reporting of cases is irregular across time and jurisdictions, leading to large deviations around a given trend. Different variants exhibit distinct temporal profiles and spatial diffusion patterns, and their contributions to total incidence evolve. The problem we address, therefore, is to construct a forecasting framework that can accommodate noisy and unevenly reported multi-variant case data and (iii) exploit transportation-based connectivity between regions to improve multi-step-ahead predictions of total and variant-specific case counts at the state level. The ultimate aim is to produce forecasts that are both statistically accurate and structurally consistent with the underlying spatial interaction processes driving disease spread.

## 3 Methodology

### 3.1 Graph Construction

Each of the following non-baseline methods predicts COVID-19 cases at the state level and incorporates a weighted state-level graph, representing the relative connections between each state with respect to transmissibility. We use three components to construct an aggregate edge weighting scheme: highway connections, airport connections, and direct border connections.

Formally, we can represent the graph  $G = (S, E)$ , where  $S$  is the set of all U.S. states, excluding Hawaii and Alaska, and including the District of Columbia, and  $E$  is the set of edge weights between

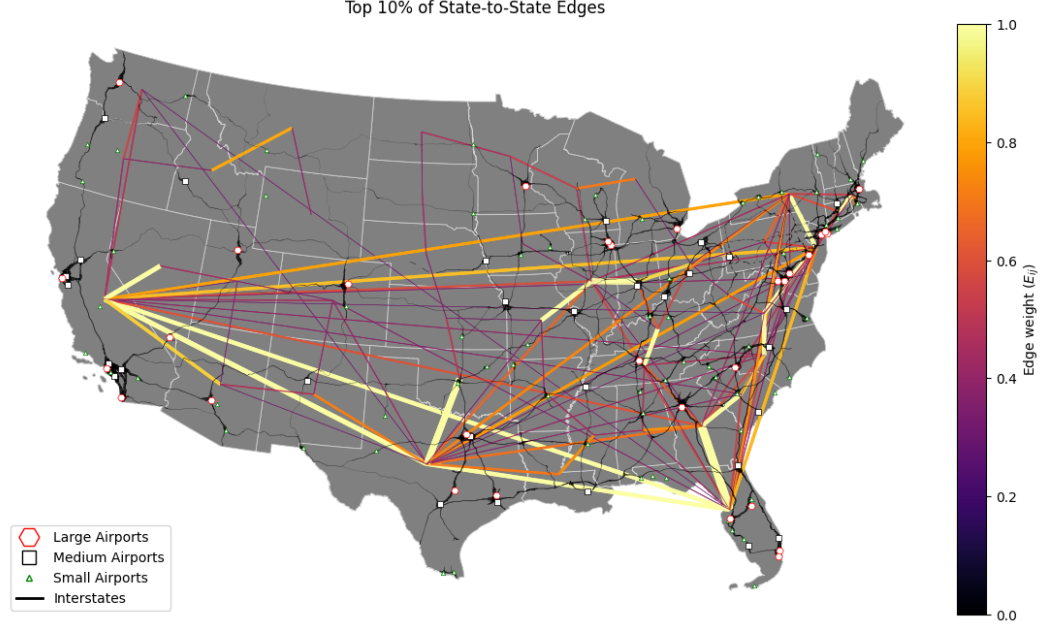


Figure 1: Selected major pairwise edges

each state. We define each edge weight as

$$E_{ij} = W_{i,j}^H + W_{i,j}^A + W_{i,j}^B$$

where the sum is of the pairwise weights for the highway, airport, and border components, respectively.

The highway weights are constructed as

$$W_{i,j}^H = \frac{M_i M_j}{C_{i,j}^2}$$

where  $M_i$  is the mass of each state, and calculated as

$$M_i = \sum_h a_h w_h d_h$$

where  $h$  denotes an interstate highway segment,  $a_h$  is the average daily traffic along a given segment,  $w_h$  is the width of that segment in lanes, and  $d_h$  is the distance in miles of a given segment. Intuitively, the mass term represents a measure of the total volume of traffic within that state.

The denominator term,  $C_{i,j}$ , is the travel time between the centroids of each pair of states. One may recognize this formulation as a variation of the commonly used gravity model, named after Newton's formula for gravity in physics, but is also commonly used in fields like trade economics.

The airport weights are constructed as

$$W_{i,j}^A = P_i P_j$$

where  $P_i$  is the total enplanements (passengers boarding flights) of all airports within 20 miles of state  $i$ . The 20-mile radius is added to account for metropolitan area effects, i.e., airports that serve cross-state metropolitan areas, such as Newark (located in New Jersey), primarily serving New York City (located in New York state).

Finally, the border weight is just the length in meters of the shared border between two states.

All weights are normalized and summed as shown earlier. Figure 1 shows the largest 10% of edges.

An intuitive check here confirms the validity of the graph construction model. For example, there are a significant amount of weights along the East Coast of the U.S., representing the I-95 corridor, a

major interstate highway. States that are coterminous with major metropolitan areas also see strong connections with each other, such as the connections between New York and New Jersey (New York City), or Illinois and Missouri (St. Louis). States that are major travel destinations also see strong connections among each other, and with states that have large wealth populations. Examples here include California and Florida connecting to New Jersey, Illinois, and Texas. Finally, small states only have strong connections to their direct neighbors, as seen in the Pacific Northwest states.

### 3.2 Baseline Methods

In this section, we introduce the baseline models used for comparison in our forecasting study. We begin with two naive temporal baselines that serve as simple but informative lower bounds. We then describe two graph-based spatio-temporal neural network architectures. Finally, we summarize the empirical performance of all baselines across forecasting horizons.

#### 3.2.1 Naive Baselines

**Last-Day Prediction.** The first baseline simply repeats the most recently observed value for all forecasting horizons. For a state  $s$  at time  $t$ , with total case counts aggregated across all variants, the prediction is

$$\hat{y}_{s,t+k} = \sum_v y_{s,t,v}.$$

This naive baseline requires no parameters and reflects the persistence assumption commonly used in epidemiological nowcasting.

**Moving Average Prediction.** The second baseline smooths short-term fluctuations by using a  $P$ -day moving average:

$$\hat{y}_{s,t+k} = \frac{1}{P} \sum_{i=t-P+1}^t \sum_v y_{s,i,v}.$$

Compared to Last-Day, the moving average baseline is more robust to day-to-day noise and tends to yield more stable performance on mid-term horizons.

#### 3.2.2 Graph-Based Baselines

**StemGNN.** The Spatio-Temporal Enhanced Multi-graph Neural Network (StemGNN) [2] is a spectral-domain forecasting model designed for multivariate time series on graphs. Instead of relying on message passing in the node domain, StemGNN jointly captures spatial and temporal dependencies by transforming signals into the frequency domain.

Given an adjacency matrix  $A$ , let  $U$  denote the normalized graph Laplacian eigenvectors. For a time series  $X \in \mathbb{R}^{T \times N}$ , StemGNN applies the Graph Fourier Transform (GFT):

$$X_{\text{GFT}} = U^\top X,$$

which maps spatial correlations into spectral coefficients. A temporal Fourier transform is then applied to each spatial frequency, producing a joint spatio-temporal spectral representation. Convolution becomes element-wise multiplication in this domain, enabling efficient modeling of long-range temporal dependencies.

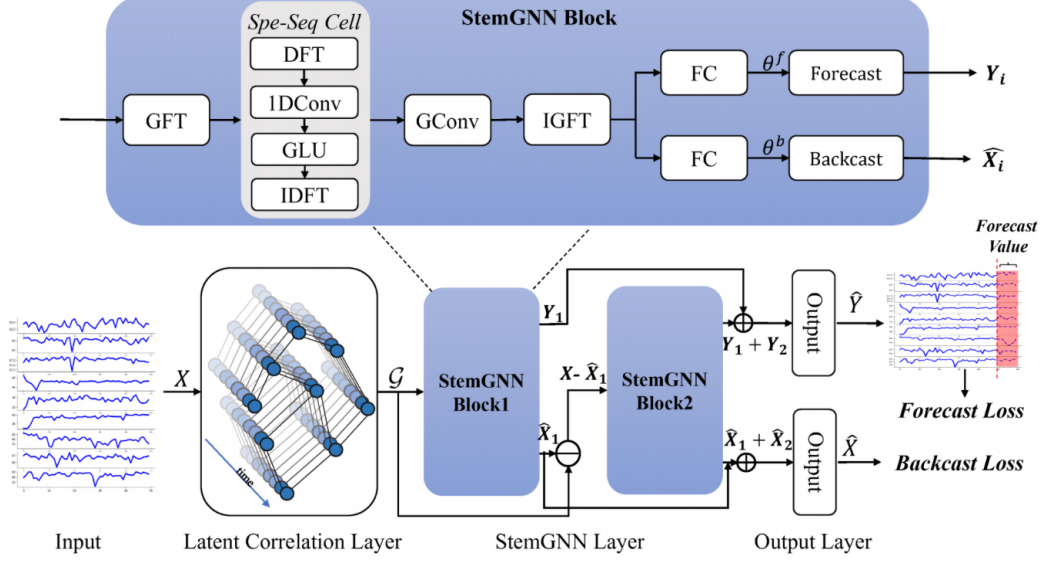


Figure 2: StemGNN structure.

StemGNN also supports multiple graphs  $\{A^{(g)}\}_{g=1}^G$ , each providing distinct structural information (e.g., geographic distance, mobility, or correlation networks). Signals are projected onto each graph’s spectral basis and fused adaptively to capture complementary relational structure.

**MTGNN.** The Multivariate Time Series Graph Neural Network (MTGNN) [8] integrates graph learning with temporal convolution. Unlike fixed-graph models, MTGNN estimates the adjacency matrix dynamically using data-driven similarity scores, allowing flexible representations even when the underlying relational structure is unknown.

The architecture consists of:

1. a graph learning module that infers directed connectivity patterns,
2. a mix-hop propagation layer that enlarges receptive fields beyond one-hop neighbors,
3. a temporal convolution network (TCN) for extracting long-range temporal dependencies.

This combination enables MTGNN to jointly model adaptive spatial relationships and temporal trends in multivariate epidemiological data.

### 3.3 StemGNN

In our work, we modify the architecture of StemGNN (described in detail previously) in two key ways. First, the original StemGNN processes only the target time series that are to be forecasted, disregarding additional covariates that may provide useful contextual information. To address this limitation, we augment the input by incorporating supplementary feature channels together with the original target series  $X$ . This enriched input representation allows the model to leverage exogenous signals that may improve forecasting performance.

Second, we re-examine the temporal modeling component used in StemGNN. After applying the Graph Fourier Transform (GFT), the original design applies a Discrete Fourier Transform (DFT) followed by a Gated Linear Unit (GLU) and then processes each resulting univariate frequency component using a Seq2Seq cell. However, once the GFT is applied, the transformed signals become approximately independent univariate time series, for which the DFT+GLU+Seq2Seq pipeline is not necessarily the most effective modeling choice. To better capture temporal dependencies within each univariate series, we replace the Seq2Seq cell with a lightweight attention block. This

modification improves the model’s ability to learn long-range temporal patterns and adapt to complex frequency-domain dynamics.

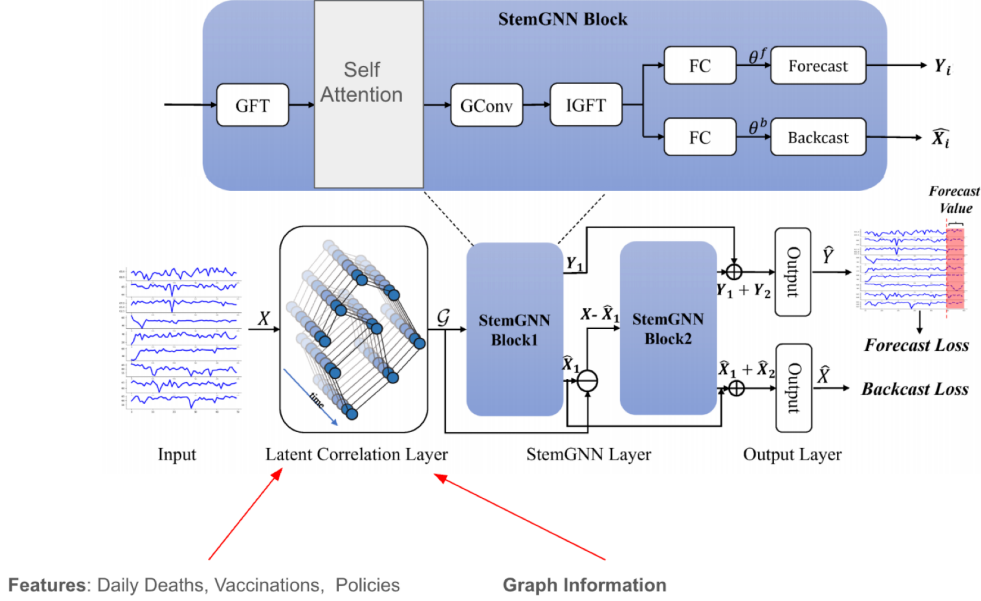


Figure 3: Modified StemGNN structure.

### 3.4 Hybrid GCN

The Hybrid GCN (HGCN) model is an approach that is designed to take advantage of the nature of COVID-19. Instead of directly predicting total cases over the future horizon, HGCN predicts *residuals of the moving average for each variant*<sup>1</sup> of COVID-19. This is because the moving average baseline is already strong (as will be discussed later). HGCN then only has to focus on the deviations from this average, which makes it well-suited for predicting data that has an underlying trend but experiences large variation from day-to-day, as can be seen in Figure 4.

<sup>1</sup>A variant here refers to a WHO-listed family of sequences. These are the commonly known Greek names like Delta, Omicron, and Iota.

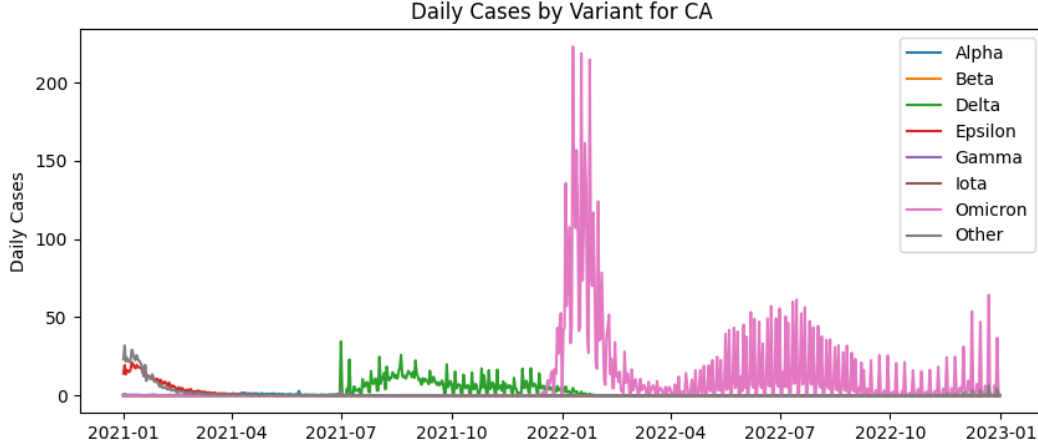


Figure 4: Daily cases by variant in the state of California

In addition, Figure 4 also illustrates the second key reason for this change; each variant has its own trends. Aggregation into total cases ignores the underlying mechanics of each variant; for example, in early 2022, the Delta variant was decreasing, while the Omicron variant was increasing.

Formally, let  $Y$  be a  $V \times S \times T$  time-series, representing the number of cases per state for each variant per day. Let  $G$  represent the state-level graph as explained earlier. We pass  $G$  into a GCN to get the hidden representations for each state. We then concatenate these representations with  $Y'_s$ , the *residuals of the moving average of the previous seven days* for each state, and pass them into a gated recurrent unit (GRU). These steps are largely inspired by the Multi-scale Spatiotemporal Neural Network [23]. After the GRU makes its predictions, we add these predictions to the moving average and calculate the loss using mean-squared error. Figure 5 shows a diagram of the whole process.

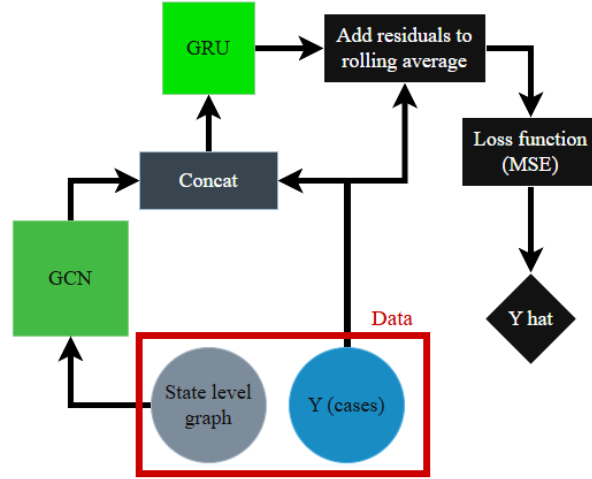


Figure 5: HGCN model structure

### 3.5 DCGRNN

The Diffusion Convolutional Gated Recurrent Neural Network (DCGRNN) is a deep learning architecture that combines temporal dynamics with spatial diffusion processes for spatio-temporal forecasting. Unlike spectral approaches, DCGRNN models spatial dependencies through a diffusion process on the graph, which naturally captures multi-hop neighborhood influence.

### 3.5.1 Diffusion Convolution

Given a graph with adjacency matrix  $A$ , the diffusion convolutional layer models information propagation as a random walk process. For a signal  $\mathbf{x} \in \mathbb{R}^N$  on the graph, the diffusion convolution is defined as:

$$\mathbf{h} = \sum_{k=0}^K \theta_k (D_O^{-1} A)^k \mathbf{x} + \sum_{k=0}^K \theta'_k (D_I^{-1} A^\top)^k \mathbf{x}$$

where  $D_O$  and  $D_I$  are the out-degree and in-degree diagonal matrices,  $K$  is the maximum diffusion step, and  $\{\theta_k, \theta'_k\}$  are learnable parameters. This formulation captures both forward and backward information diffusion across the transportation network.

### 3.5.2 Encoder-Decoder Architecture

DCGRNN employs a sequence-to-sequence architecture with GRU cells augmented by diffusion convolution. The encoder processes historical observations  $\{y_{t-T+1}, \dots, y_t\}$  to produce hidden states, which are then passed to a decoder that autoregressively generates future predictions  $\{\hat{y}_{t+1}, \dots, \hat{y}_{t+H}\}$  over the forecast horizon  $H$ .

At each time step, the GRU update and reset gates are replaced with diffusion convolutional operations:

$$\begin{aligned} r_t &= \sigma(\text{DiffConv}([\mathbf{x}_t, h_{t-1}], A) + b_r) \\ u_t &= \sigma(\text{DiffConv}([\mathbf{x}_t, h_{t-1}], A) + b_u) \\ \tilde{h}_t &= \tanh(\text{DiffConv}([\mathbf{x}_t, r_t \odot h_{t-1}], A) + b_h) \\ h_t &= u_t \odot h_{t-1} + (1 - u_t) \odot \tilde{h}_t \end{aligned}$$

where  $\odot$  denotes element-wise multiplication and  $\text{DiffConv}(\cdot, A)$  represents the diffusion convolution operation.

## 4 Experiments

### 4.1 Datasets

Our models rely on seven different datasets for our experiments. To construct our covariates dataset, we use **variant data** from GISAID [4], accessed with the Outbreak.info R package [1], and combined with New York Times data on **daily cases**, accessed via the COVID-19 Data Hub [9]. Our data on **exogenous covariates** also came from the COVID-19 Data Hub. These are primarily data on the relative severity of NPIs (non-pharmaceutical interventions) such as school/restaurant closures, travel restrictions, mask mandates, and aggregate indices of these policies. Other data included are population and density statistics. All covariate and cases data are from January 1, 2021, to January 1, 2023.

To construct the state-level graph, we used **state-level shapefile data** from the U.S. Census Bureau [20], **highway shapefile data** from the U.S. Department of Transportation [21], **airport location and enplanement data** from the Federal Aviation Administration [5] [6], and **county-level adjacency data** from the U.S. Census Bureau [19].

### 4.2 Experimental Setup

First, to determine the appropriate baseline to compare our models against, we test each of the baselines on a one to seven-day prediction horizon, as well as their average performance over the total horizon. We test each baseline on three metrics: mean absolute percent error (MAPE), root mean squared error (RMSE), and mean absolute error (MAE).

Once we select the highest performing baseline, we test each of the three non-baseline models on the same horizon and with the same metrics.



### 4.3 Results

The experimental results indicate that the highest performing baseline is the moving average, as indicated in Table 1. This baseline is only slightly better than the last-day baseline in terms of MAPE, but is significantly better on RMSE and MAE than any other baseline.

Table 1: Results for each baseline over the seven-day horizon

| Metric | Model     | H=1            | H=2            | H=3            | H=4            | H=5            | H=6            | H=7            | Avg            |
|--------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| MAPE   | StemGNN   | 1.0676         | 1.8843         | 1.6147         | 1.9031         | 1.5162         | 1.7747         | 1.3762         | 1.5910         |
|        | StemGNN_W | 1.0776         | 1.8844         | 1.5896         | 1.8996         | 1.5347         | 1.7685         | 1.3735         | 1.5897         |
|        | MTGNN     | 0.7657         | 2.2407         | 2.2425         | 2.2389         | 2.2324         | 2.2280         | 2.2311         | 2.0256         |
|        | MTGNN_W   | <b>0.7510</b>  | 2.2407         | 2.2425         | 2.2389         | 2.2324         | 2.2280         | 2.2311         | 2.0235         |
|        | Last-Day  | 1.0149         | 0.9662         | 1.0130         | 1.0155         | 0.9909         | 0.9546         | <b>0.4254</b>  | 0.9115         |
|        | MovingAvg | 0.9093         | <b>0.9071</b>  | <b>0.9062</b>  | <b>0.9067</b>  | <b>0.9038</b>  | <b>0.9042</b>  | 0.9030         | <b>0.9058</b>  |
| RMSE   | StemGNN   | 2930.82        | 4513.43        | 4054.01        | 4722.15        | 3862.28        | 4419.07        | 3639.32        | 4020.15        |
|        | StemGNN_W | 2937.09        | 4526.22        | 3977.99        | 4734.25        | 3883.43        | 4410.45        | 3627.88        | 4013.90        |
|        | MTGNN     | <b>2644.75</b> | 3564.70        | 3831.36        | 3815.54        | 3795.81        | 3795.63        | 3795.66        | 3606.21        |
|        | MTGNN_W   | 2645.99        | 3564.70        | 3831.36        | 3815.54        | 3795.81        | 3795.63        | 3795.66        | 3606.38        |
|        | Last-Day  | 4480.64        | 4580.82        | 4476.72        | 4486.12        | 4537.15        | 4469.88        | <b>2174.01</b> | 4172.19        |
|        | MovingAvg | 3089.70        | <b>3078.83</b> | <b>3090.44</b> | <b>3106.34</b> | <b>3125.29</b> | <b>3144.24</b> | 3165.23        | <b>3113.63</b> |
| MAE    | StemGNN   | 1405.92        | 2914.68        | 2464.33        | 3013.20        | 2286.82        | 2773.10        | 2057.77        | 2416.55        |
|        | StemGNN_W | 1427.40        | 2916.38        | 2407.73        | 3010.59        | 2323.46        | 2761.42        | 2048.21        | 2413.60        |
|        | MTGNN     | <b>1013.45</b> | 2545.51        | 2580.17        | 2572.93        | 2571.45        | 2571.03        | 2571.54        | 2346.58        |
|        | MTGNN_W   | 1032.45        | 2545.51        | 2580.17        | 2572.93        | 2571.45        | 2571.03        | 2571.54        | 2349.30        |
|        | Last-Day  | 1530.27        | 1594.86        | 1598.71        | 1596.00        | 1577.98        | 1507.35        | <b>458.15</b>  | 1409.05        |
|        | MovingAvg | 1279.11        | <b>1270.16</b> | <b>1271.40</b> | <b>1274.76</b> | <b>1281.89</b> | <b>1289.30</b> | 1299.47        | <b>1280.87</b> |

Testing our models against the moving average, we find that the only model to outperform it is the HGCN<sup>2</sup>. Table 2 shows that HGCN performs roughly 9-23% better than the baseline, while the other models perform significantly worse.

Table 2: Results for each model and selected baseline over the seven-day horizon

| Metric | Model     | H=1            | H=2            | H=3            | H=4            | H=5            | H=6            | H=7            | Avg            | % Under MA    |
|--------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|
| MAPE   | MovingAvg | 0.9093         | <b>0.9071</b>  | 0.9062         | 0.9067         | 0.9038         | 0.9042         | 0.9030         | 0.9058         | -             |
|        | HGCN      | <b>0.8345</b>  | 0.9656         | <b>0.7946</b>  | <b>0.9000</b>  | <b>0.8915</b>  | <b>0.7184</b>  | <b>0.6512</b>  | <b>0.8212</b>  | <b>9.34%</b>  |
|        | StemGNN_R | 1.4584         | 1.7018         | 1.2760         | 1.0568         | 1.1516         | 1.8414         | 1.5832         | 1.438          | -58.81%       |
|        | DCGRNN    | 1.4498         | 1.2862         | 1.3798         | 1.5830         | 1.6576         | 1.6529         | 1.7680         | 1.5396         | -69.97%       |
| RMSE   | MovingAvg | 3089.70        | 3078.83        | 3090.44        | 3106.34        | 3125.29        | 3144.24        | 3165.23        | 3113.63        | -             |
|        | HGCN      | <b>2297.65</b> | <b>2713.47</b> | <b>2526.24</b> | <b>2457.98</b> | <b>2343.50</b> | <b>2237.08</b> | <b>2265.41</b> | <b>2411.11</b> | <b>22.56%</b> |
|        | StemGNN_R | 3634.50        | 4229.53        | 3496.77        | 3309.49        | 3432.45        | 4630.62        | 4133.28        | 3838.09        | -23.27%       |
|        | DCGRNN    | 2799.10        | 2901.81        | 3154.05        | 3319.53        | 3257.12        | 3483.43        | 3443.98        | 3194.15        | -2.59%        |
| MAE    | MovingAvg | 1279.11        | <b>1270.16</b> | 1271.40        | 1274.76        | 1281.89        | 1289.30        | 1299.47        | 1280.87        | -             |
|        | HGCN      | <b>1018.97</b> | 1285.82        | <b>1089.05</b> | <b>1218.76</b> | <b>1158.94</b> | <b>846.266</b> | <b>818.543</b> | <b>1062.34</b> | <b>17.06%</b> |
|        | StemGNN_R | 2121.67        | 2649.14        | 1851.14        | 1465.37        | 1654.35        | 2962.36        | 2496.03        | 2171.44        | -69.53%       |
|        | DCGRNN    | 1315.03        | 1453.86        | 1591.14        | 1614.25        | 1544.03        | 1667.24        | 1658.4         | 1549.14        | -17.32%       |

Figure 6 shows HGCN’s next-day predictions against the ground truth and moving average. In each of the states, HGCN outperforms the moving average. Crucially, the model is able to pick up both the underlying trends over time, but also the intra-week variation that leads to the large spikes.

<sup>2</sup>Metrics for HGCN are on total cases derived from aggregating each individual variant prediction

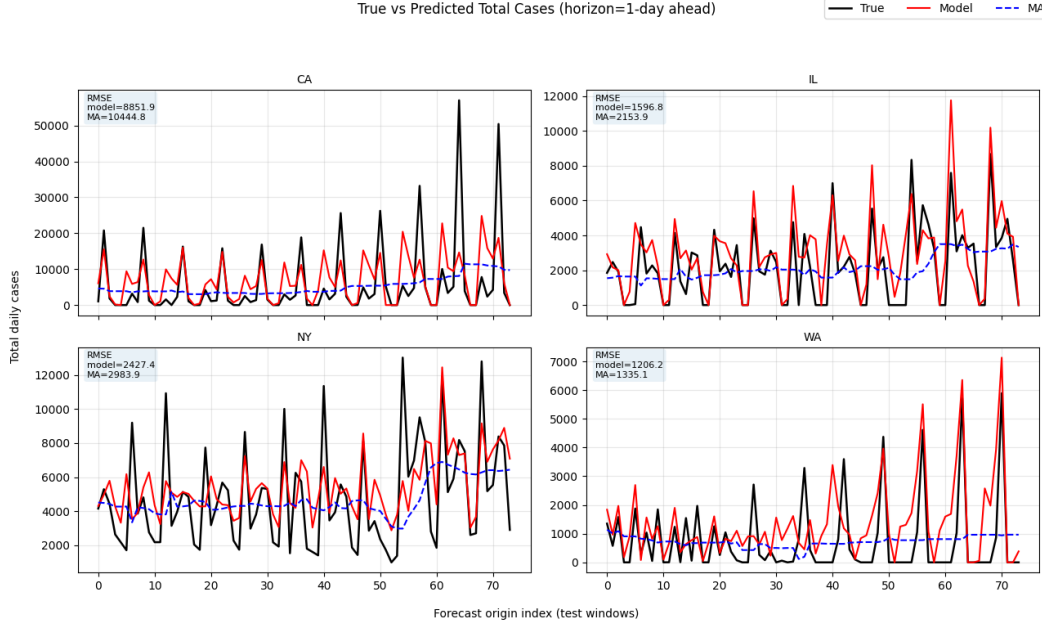


Figure 6: True versus predicted cases for four states

## 5 Conclusion and Future Work

In this work, we conducted a comprehensive study on forecasting COVID-19 case counts using a range of baseline methods, graph-based neural models, and several architectural modifications tailored to spatio-temporal dynamics. Our empirical results highlight several key observations.

First, despite their simplicity, naive baselines—particularly the moving average predictor—provide a surprisingly competitive reference point. This suggests that short-term fluctuations in reported case counts are modest relative to underlying trends, making temporal smoothing highly effective. The strong performance of the moving average baseline also indicates that the data exhibit limited short-horizon volatility, which future models should explicitly account for.

Second, graph-based models such as StemGNN, Hybrid GCN, and DCGRNN demonstrate varying degrees of improvement over the baselines. Among these methods, the Hybrid GCN achieved the most consistent performance gains, especially for short forecasting horizons. Its residual-prediction strategy—which leverages the moving average as a strong anchor—proved to be an effective way of stabilizing learning and reducing forecasting error. By contrast, more complex spectral or diffusion-based architectures offered limited benefits in this setting. This may be due to the nature of disease spread, where spatial propagation patterns are relatively stable and may already be well-captured by coarse-level mobility graphs.

Third, our experiments highlight that incorporating additional covariates (e.g., county-level demographic and socioeconomic variables) does not necessarily improve performance. In some configurations, these features introduced computational overhead with negligible forecasting gains. One possible explanation is that structural information encoded by transportation networks (e.g., airport traffic, highways) already captures a significant portion of the spatial heterogeneity relevant for transmission dynamics.

Taken together, these findings suggest that the forecasting task is dominated by stable temporal trends with modest spatial variation, making simple models harder to outperform than expected. They also demonstrate that spatial information can indeed improve performance, but such improvements depend heavily on how the temporal component is modeled.

## Future Work

There are several promising directions for extending this work:

- **Improved handling of reporting irregularities.** Daily case counts often exhibit administrative noise or periodic reporting delays. Future models could incorporate explicit de-biasing or temporal calibration modules to mitigate these effects.
- **Dynamic graph structures.** Our current models assume static mobility or contact networks. Incorporating time-varying graphs—for example, based on real-time mobility data or evolving transportation patterns—may allow better modeling of transmission dynamics.
- **Probabilistic and uncertainty-aware forecasting.** Disease forecasting inherently involves substantial uncertainty. Extending the current models with Bayesian methods, uncertainty quantification, or ensemble techniques could improve robustness and interpretability.
- **Multi-resolution forecasting.** Our results show meaningful differences between daily and weekly predictions, suggesting that multi-scale temporal modeling could be beneficial.
- **Integration of variant-specific dynamics.** Models that explicitly account for heterogeneous behavior across variants may provide more accurate forecasts, particularly during periods of rapid variant replacement.

Overall, while simple baselines remain surprisingly strong, incorporating spatial structure and improved temporal modeling holds substantial promise for advancing the accuracy of epidemic forecasting models.

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